

CLIMATE CHANGE: THE PACIFIC CATCH 22

Rising sea levels, ocean warming, and ocean acidification threaten the socioeconomic stability and existence of Pacific Islands. International tourism is relied upon to fund climate adaptation, however significant increases in tourism undermine decarbonisation aspirations. This review considers the dynamics between the strain of tourism and renewable capacity, and opportunities for Pacific nations to work together to reduce reliance on carbon fuels.

Project by Kingsley Beagle, August 2026

Renewable energy share has suffered, while tourism has soared

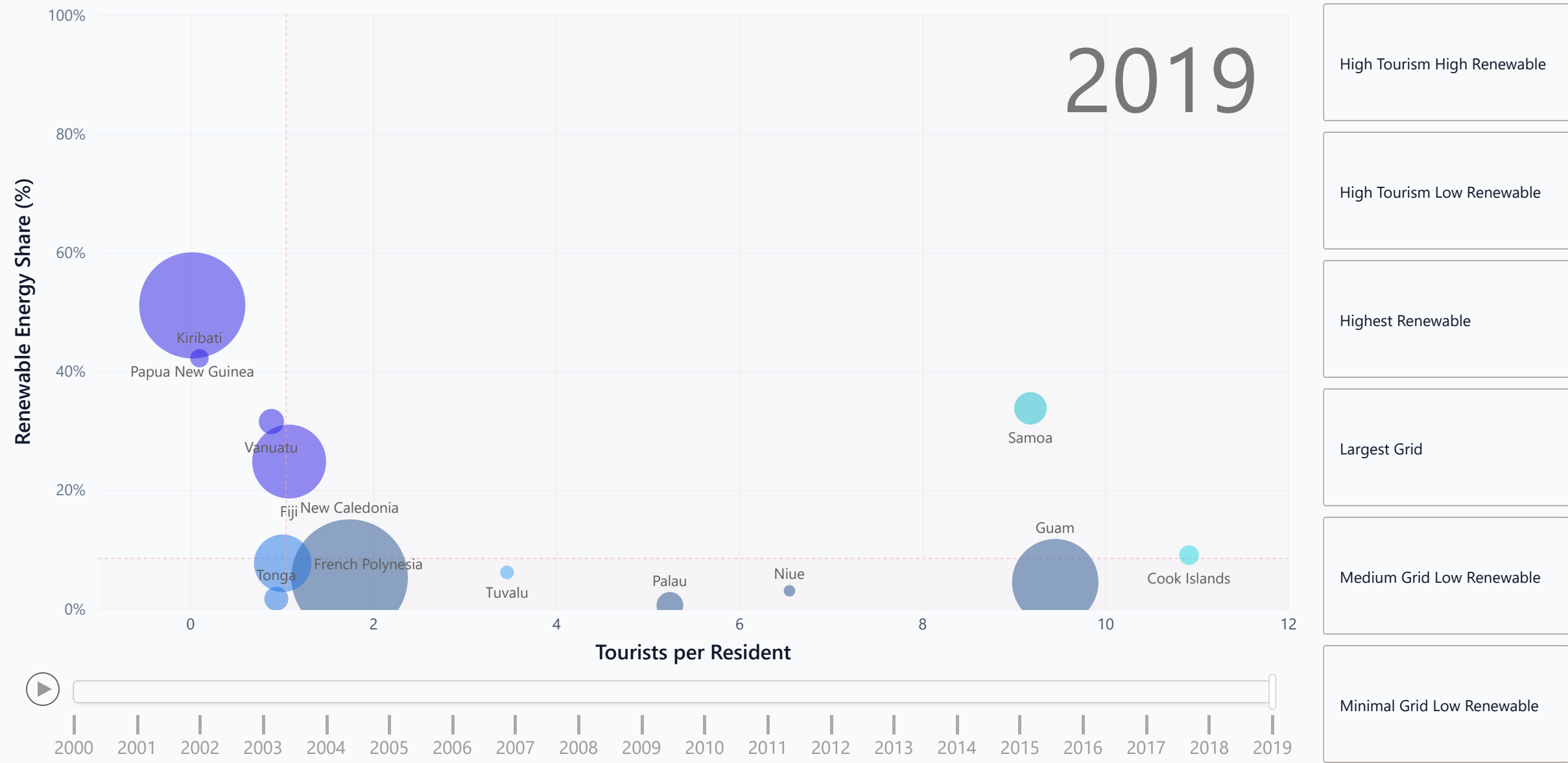
Red dashed lines indicate median values for Tourists per Resident, and Renewable Energy Share (%) for the year 2000. Locations which started below the median **failed to reduce their reliance on carbon fuel sources**, including New Caledonia, who had one of the largest expansions. Renewable pioneers from 2000 (Papua New Guinea, Kiribati, Vanuatu, Fiji, and Samoa) all experienced a **reduction in Renewable Energy Share**.

The economic boost from **tourism has not been enough to displace energy demands** placed on the islands by those very same visitors. Local tourism as a source of economic resilience inadvertently locks Pacific Island nations into a state of fossil fuel dependency.

Pacific Island nations are clustered algorithmically based on relevant features, independent of land area. Palau and Niue are the only locations which transition between clusters, due to significant increases in tourism.

To create reliable clusters, Pacific Island nations with missing data were omitted, and the largest possible period of consecutive data was selected. The time range was intentionally capped at 2019, to ensure the reduction in tourism from COVID-19 did not obscure the analysis.

Tourism grows (bubbles move right), renewable performance decreases or stagnates (bubbles never rise), and energy demands vary (bubble sizes change).
Selecting a bubble illustrates local impact.
Selecting a cluster (right of chart) illustrates shared behaviours.



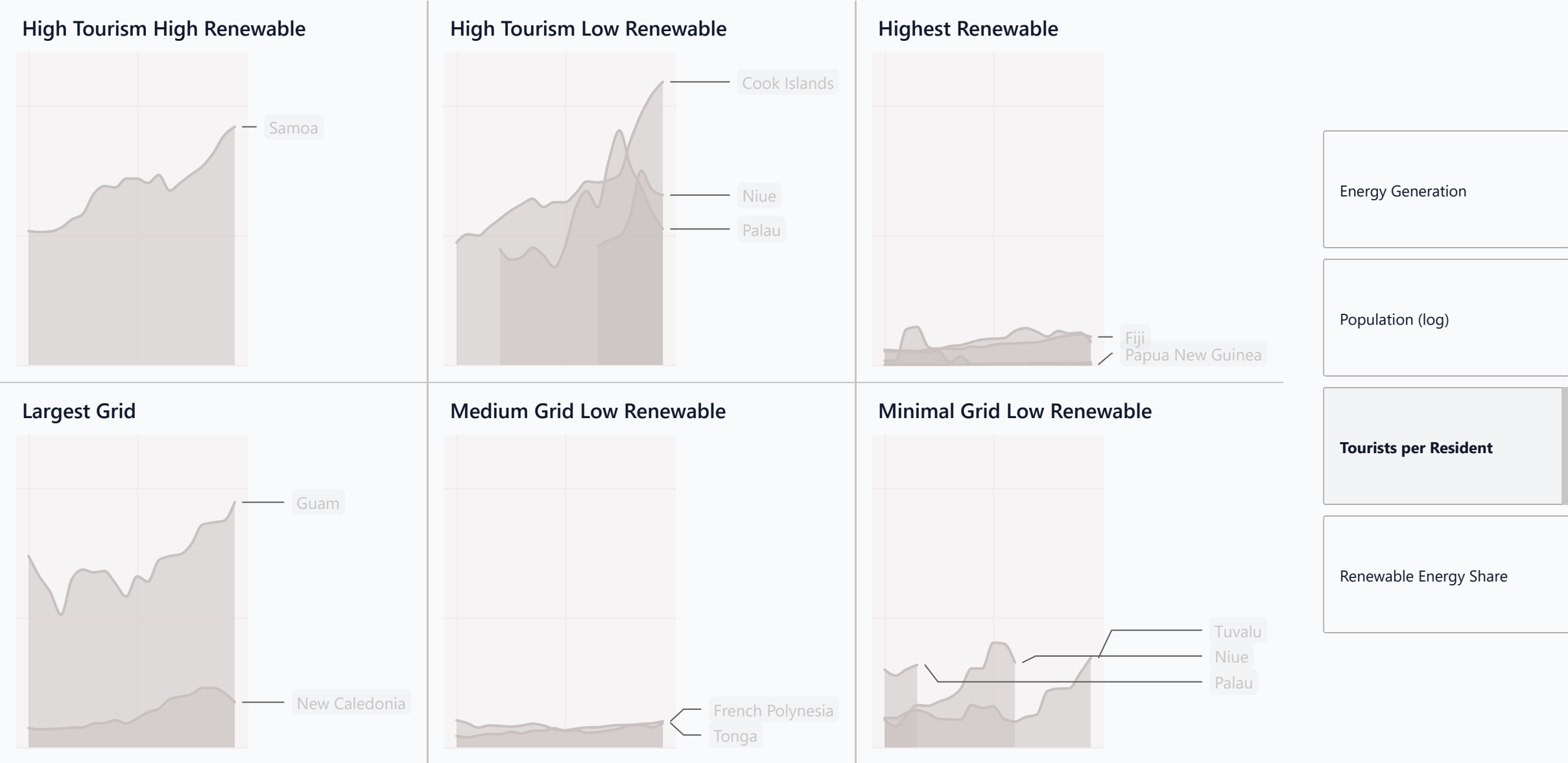
Cluster profiles

The line charts show each of the clusters, while the buttons to the right represent the input features used for clustering. Each chart uses a common scale - values and units are intentionally omitted. These charts are designed to help the user explore similarities across Pacific Island locations within clusters - they are not intended for statistical deep dive.

Clusters with **high starting grid capacity have scale advantage**, their large grids absorb tourist load because of existing capability. For these states the transition toward a higher renewable share is a capital expenditure problem.

Clusters containing **small scale grids suffer from spatial restrictions** paired with high tourist to resident ratios. Land scarcity limits solar opportunity, resulting in high usage of liquid fuel supply.

Features (right of chart) dynamically illustrate how locations compare against one another across constraints and clusters.

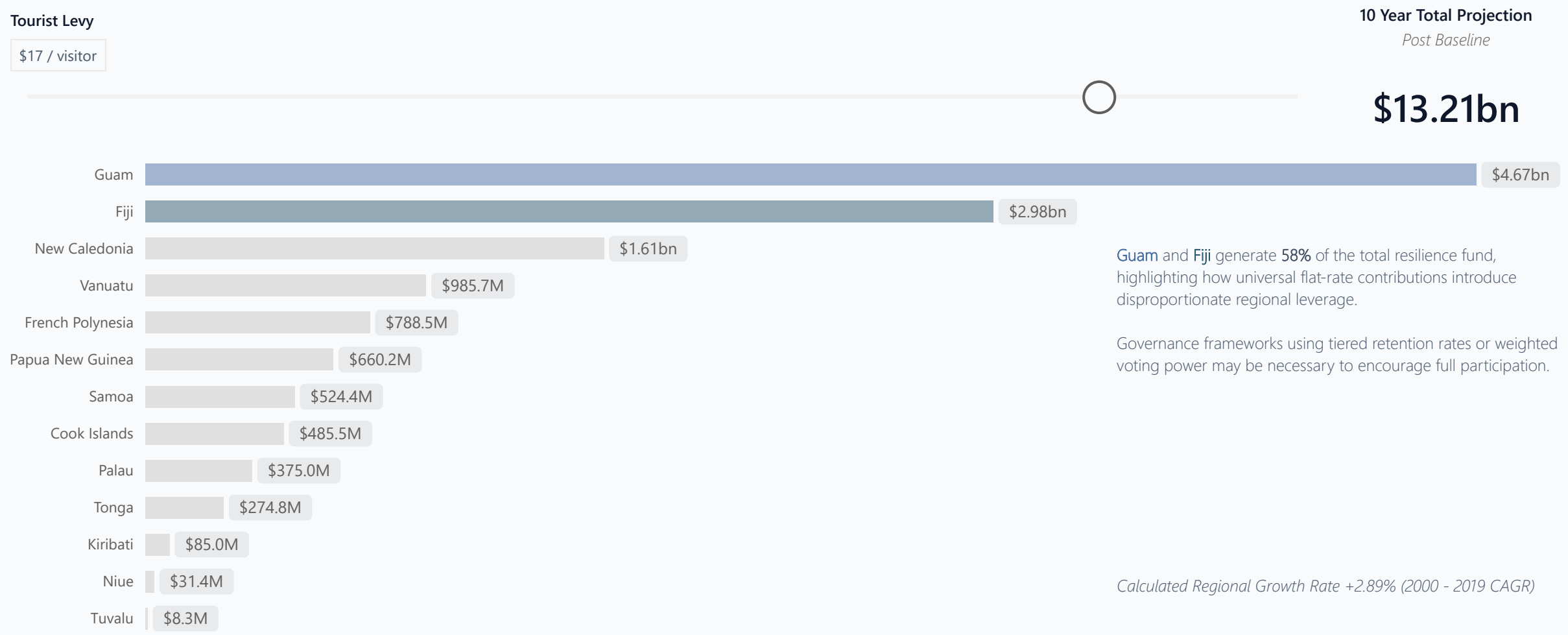


Data-driven regional policy and capital distribution opportunities

Having considered the behaviour and underlying structural drivers of individual clusters, we now focus on how clusters can unite to combat climate change. This simulator demonstrates the opportunity in a **Shared Pacific Resilience Fund**, driven by a modest tourist levy (\$1 - \$20 / visitor).

Pre-pandemic baseline tourist growth (+2.89% CAGR) is leveraged to forecast visitor numbers over a ten-year horizon. The simulation illustrates how modest visitor contributions in high tourist areas can be aggregated into a shared fund, to unlock **equitable renewable energy capital opportunities that leverage economies of scale**.

Crucially, this suggestion is not intended to be an additional tax on top of existing island fees, instead it demonstrates how data modelling can support tariff restructuring and ring fencing decisions. When individual Pacific Island nations rely on tourism to fund their progress towards decarbonisation the local touristic burden outweighs the opportunity. With this approach Pacific Island nations can **turn tourism into collective climate change resilience**.



Inter-cluster

Cross subsidised battery infrastructure

A fraction of funds from high volume hubs are pooled to subsidise expensive utility scale battery reserves for vulnerable micro-grid locations.

Targeted grid resilience grants

Revenue is routed directly to islands hit hardest by diesel volatility and space constraints - reducing upfront hardware costs without taxing local communities.

Intra-cluster

Joint regional procurement

Clusters with similar infrastructure profiles bundle their technology demands into single, large scale projects and purchases to negotiate utility-scale pricing from global solar and battery manufacturers.

Harmonised standards

Regional peers adopt unified technical grid standards, removing redundant regulatory barriers and accelerating clean deployments across islands.

Conclusion

This project successfully demonstrated how using open data to identify shared opportunity and instruct policy can transform common challenges into shared opportunity. With dynamic modelling decision-makers are able to evaluate potential inter-cluster equity solutions, alongside intra-cluster peer efficiencies.

Limitations:
This framework serves as a proof of concept for data-driven policy, and should not be considered as an accurate forecast.
Limitations include; pre-2020 data range, missing non-reporting nations, assumed universal participation, local space constraints and land availability.

Method:
Data was exported from the [Pacific Data Hub](#) and cleaned using python.
Tourism data was aggregated, and filtering was applied to ensure common locations and a consistent time frame.
Historical population estimates were constructed via back-casting.
All data was then merged into a single data frame.
K-Means clustering was used to identify distinct clusters, using the elbow method to determine the optimal number of clusters.
Clustering used *Log Population*, *Tourism per Capita*, *Energy per Capita*, and *Renewable Energy Share* as input features.
Power BI was used to generate all visualisations in this report.
Google Gemini was used to iterate on ideas and concepts from end to end, including creation of the JSON theme.

Pacific Data Hub sources used:
[Renewable energy share in the total final energy consumption](#)
[Power generation](#)
[Tourism arrivals - disaggregated](#)
[Population growth](#)
[Population used for back-casting](#)